Program 7 : Deploy Web Application to Kubernetes using Minikube

file:

|

|------nginx-deployment.yaml

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$
```

Steps:

1) minikube start

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ minikube start
😄  minikube v1.37.0 on Ubuntu 22.04
    ▪ KUBECONFIG=/home/1RV24MC108_SWASTHIK_DEVADIGA/.kube/config
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🤷  docker "minikube" container is missing, will recreate.
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...

🧯  Docker is nearly out of disk space, which may cause deployments to fail! (98% of capacity). You can pass '--force' to skip this check.
💡  Suggestion:

    Try one or more of the following to free up space on the device:

    1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
    2. Increase the storage allocated to Docker for Desktop by clicking on:
    Docker icon > Preferences > Resources > Disk Image Size
    3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🍿  Related issue: https://github.com/kubernetes/minikube/issues/9024


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🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
```

2) apply nginx-deployment.yaml and check status and expose the deployment service

```
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl apply -f nginx-deployment.yaml
loyment.apps/nginx-deployment configured
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get deployments
ME                READY   UP-TO-DATE   AVAILABLE   AGE
inx-deployment    2/2     2            19h         19h
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get pods
ME                              READY    STATUS     RESTARTS       AGE
inx-deployment-6f966446b-s1gf 1/1      Running    2 (66s ago)  33n
inx-deployment-6f966446b-d5ngw1/1      Running    2 (66s ago)  33n
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
vice/nginx-deployment exposed
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get services
ME                TYPE      CLUSTER-IP   PORT(S)  EXTERNAL-IP  PORT(S)       AGE
bernetes          ClusterIP 10.96.0.1    <none>   <none>       443/TCP       23d
be                NodePort  10.110.47.132         <none>       8888:32191/TCP 23d
inx-deployment    NodePort  10.110.47.200         <none>       80:30331/TCP   11s
inx-service       NodePort  10.110.55.196         <none>       80:30066/TCP   19h
24MC100_SHASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ minikube service nginx-deployment
```

JAMESPACE	NAME	TARGET PORT	URL
lefault	nginx-deployment	80	http://192.168.49.2:30331

```
Opening service default/nginx-deployment in default browser...
```

Step 11: Delete all the pods

Step 12: Delete all the services

Step 13: Delete al the deployments